H-Infinity Lateral Steering Control Design for a Linear Bicycle Vehicle Model

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Abstract

This report presents the derivation, formulation, and simulation evaluation of an H_∞ lateral steering controller for a constant-speed ground vehicle. The plant is modeled as a linear bicycle model with lateral position, yaw angle, lateral velocity, and yaw rate states. A side-wind-speed input is added by linearizing the lateral aerodynamic force and applying it at the vehicle center of gravity. The controller is synthesized in MATLAB using named-signal block interconnection and `hinfsyn`. The weighted generalized plant includes lateral tracking-error performance, steering command shaping, side-wind disturbance shaping, and optional measurement-noise channels. The final closed-loop analysis is performed with an unweighted augmented plant and linear fractional transformation (LFT), so the reported responses correspond to the physical lateral plant rather than the weighted synthesis model.

1 Introduction

Automatic steering can be formulated as a robust lateral tracking problem in which the steering input must regulate vehicle lateral position and heading in the presence of disturbances and modeling uncertainty. Rafaila and Livint studied a related H_∞ automatic steering problem for a vehicle moving at constant speed, using loop-shaping ideas and closed-loop simulation to evaluate steering performance [1]. This project follows the same constant-speed steering setting, but implements the full MATLAB design workflow for the current AlpaSim-style lateral model.

The main contributions of this project are:

- derivation of a lateral-only state-space bicycle model using the rear/rig lateral position as the tracked position;
- inclusion of a signed side-wind-speed disturbance by local aerodynamic-force linearization;
- construction of an H_∞ generalized plant using MATLAB named signals and `connect`;
- construction of clean and noisy closed-loop analysis models using `lft` around the original unweighted plant;
- implementation of reusable time-domain tracking simulations for trajectory tracking and wind-disturbance rejection.

2 Vehicle Lateral Dynamics

2.1 Modeling Assumptions

The model is a lateral-only linear bicycle model. The longitudinal speed is held constant at

$$v_x = 10 \text{ m/s}. \tag{1}$$

The small-angle and small-slip assumptions are used, tire forces are assumed linear in slip angle, and longitudinal, roll, tire-saturation, and hard actuator saturation effects are neglected. The steering command is applied directly as the front steering angle. Side wind is modeled as a lateral force acting at the center of gravity (CG), so it has no direct yaw moment.

The lateral state vector is

$$x = \begin{bmatrix} y_{\text{rig}} & \psi & v_y & r \end{bmatrix}^T, \tag{2}$$

where y_{rig} is the lateral position of the rear/rig reference point, ψ is yaw angle, v_y is lateral velocity at the CG, and r is yaw rate. The plant input vector is

$$u = \begin{bmatrix} \delta & v_w \end{bmatrix}^T, \tag{3}$$

where δ is steering command and v_w is signed side-wind speed.

2.2 Linear Tire Forces

Let l_f and l_r be the distances from the CG to the front and rear reference points, and let C_{af} and C_{ar} be the front and rear cornering stiffnesses. The linearized front and rear lateral tire forces are

$$F_{yf} = 2C_{af}\left(\delta - \frac{v_y + l_f r}{v_x}\right), \tag{4}$$

$$F_{yr} = 2C_{ar}\left(-\frac{v_y - l_r r}{v_x}\right). \tag{5}$$

The lateral and yaw equations of motion are

$$m\left(\dot{v}_y + v_x r\right) = F_{yf} + F_{yr} + F_w, \tag{6}$$

$$I_z \dot{r} = l_f F_{yf} - l_r F_{yr}. \tag{7}$$

2.3 Side-Wind Linearization

The nonlinear aerodynamic side force is modeled as

$$F_w = \frac{1}{2}\rho C_y A_s |v_w| v_w, \tag{8}$$

where ρ is air density, C_y is the side-force coefficient, and A_s is the side projected area. Around a nonzero operating wind speed v_{w0}, the local linearization is

$$\Delta F_w \approx k_w \Delta v_w, \qquad k_w = \rho C_y A_s |v_{w0}|. \tag{9}$$

Since the force is applied at the CG, the direct yaw moment from wind is zero. Thus the wind input enters only through the lateral acceleration equation.

2.4 State-Space Form

The kinematics of the rear/rig reference point are

$$\dot{y}_{\text{rig}} = v_x \psi + v_y - l_r r, \tag{10}$$

$$\dot{\psi} = r. \tag{11}$$

Combining (4)–(11) gives

$$\dot{x} = Ax + Bu, \qquad y = Cx + Du, \tag{12}$$

where

$$A = \begin{bmatrix} 0 & v_x & 1 & -l_r \\ 0 & 0 & 0 & 1 \\ 0 & 0 & a_{00} & a_{01} \\ 0 & 0 & a_{10} & a_{11} \end{bmatrix}, \tag{13}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ b_{00} & k_w/m \\ b_{10} & 0 \end{bmatrix}, \tag{14}$$

and

$$a_{00} = -\frac{2(C_{af} + C_{ar})}{m v_x}, \tag{15}$$

$$a_{01} = -v_x - \frac{2(l_f C_{af} - l_r C_{ar})}{m v_x}, \tag{16}$$

$$a_{10} = -\frac{2(l_f C_{af} - l_r C_{ar})}{I_z v_x}, \tag{17}$$

$$a_{11} = -\frac{2(l_f^2 C_{af} + l_r^2 C_{ar})}{I_z v_x}, \tag{18}$$

$$b_{00} = \frac{2C_{af}}{m}, \tag{19}$$

$$b_{10} = \frac{2 l_f C_{af}}{I_z}. \tag{20}$$

The current implementation uses $C = I_4$, so all four states are available as analysis outputs.

3 Control Problem Formulation

The objective is to synthesize a steering controller that reduces lateral tracking error while rejecting side-wind disturbances. The measured lateral tracking error used by the current controller is

$$e_y = r_y - y_{\text{rig},ns}, \tag{21}$$

where r_y is the lateral reference and $y_{\text{rig},ns}$ is the possibly noisy measured lateral position.

The generalized plant input vector used for synthesis is

$$w = \begin{bmatrix} r_y & r_\psi & n_y & n_\psi & d_w \end{bmatrix}^T, \qquad u_c = \delta_{\text{in}}, \tag{22}$$

where d_w is the shaped side-wind input and u_c is the controller output. The active performance output is

$$z = W_{a_y}(s) e_y. \tag{23}$$

The controller measurement is

$$y_c = e_y. \tag{24}$$

The MATLAB command

$$[K_1, S_{\text{cl}}, \gamma] = \texttt{hinfsyn}(P, n_{\text{meas}}, n_{\text{cont}}) \tag{25}$$

solves the standard H_∞ problem

$$\min_{K_1} \|T_{zw}(s)\|_\infty, \tag{26}$$

where T_{zw} is the closed-loop transfer matrix from exogenous inputs w to weighted performance outputs z. In the current code,

$$n_{\text{meas}} = 1, \qquad n_{\text{cont}} = 1. \tag{27}$$

Therefore the final generalized-plant output is treated as the controller measurement and the final generalized-plant input is treated as the controller command.

4 Block Interconnection

4.1 Synthesis Plant

The synthesis plant, named `nominal_op` in MATLAB, is built from:

- `lateral_sys`: original physical state-space plant;
- `Rpass`: reference pass-through block;
- `Wn`: measurement-noise scaling block;
- `Wa_y`: lateral tracking-error performance weight;
- `Wa_yaw`: yaw tracking weight, currently defined but not active;
- `W_cmd`: steering command shaping block;
- `W_wind`: side-wind disturbance shaping block.

The current synthesis output list is

$$[e_{y1}, \ e_y]^T, \tag{28}$$

where $e_{y1} = W_{a_y} e_y$. Thus, only the lateral position error is currently included in the H_∞ performance objective. The yaw error is computed and plotted for analysis, but $e_{\psi 1}$ is not presently included in the synthesis output vector.

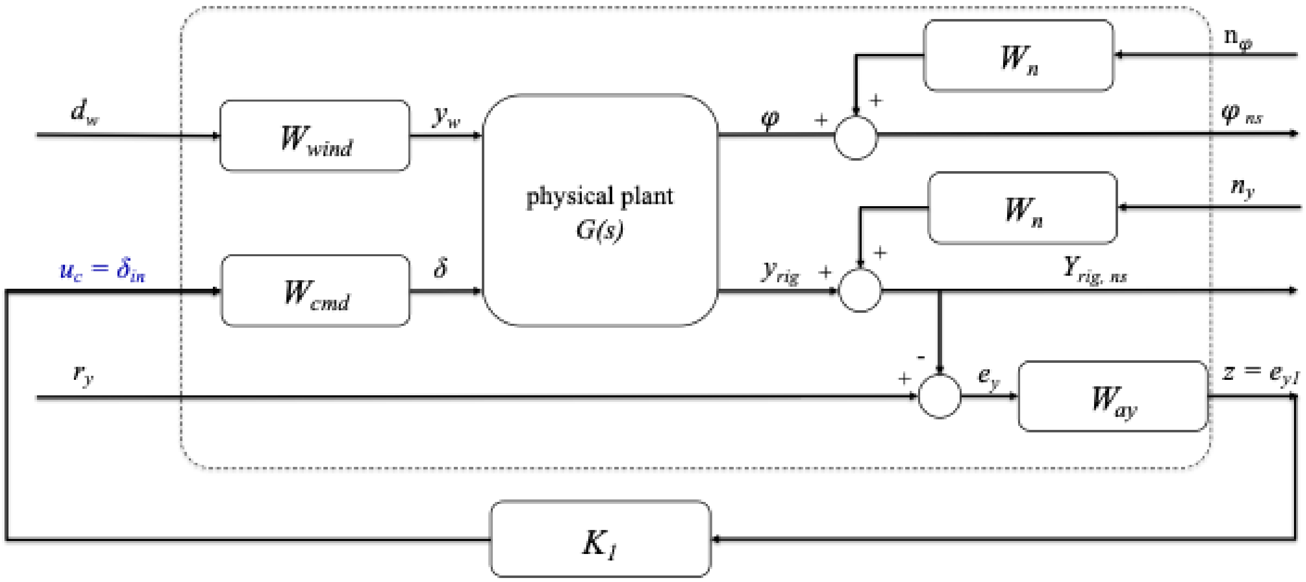

Figure 1: Generalized-plant interconnection used for H_∞ synthesis. The dashed box is $P(s)$ and includes the physical plant, weighting filters, and summing junctions. The controller K_1 is outside $P(s)$, receives the measured tracking error $y_c = e_y$, and returns the normalized steering command $u_c = \delta_{\text{in}}$.

4.2 Unweighted Closed-Loop Analysis Plant

After synthesis, the controller K_1 maps

$$e_y \mapsto \delta_{\text{in}}. \tag{29}$$

The physical plant expects the physical steering command δ, so the implemented controller used for closed-loop analysis is

$$K_{\text{plant}}(s) = W_{\text{cmd}}(s)K_1(s). \tag{30}$$

The clean analysis plant is named `nominal_augment_lateral_system`. It is built from the original unweighted plant and error-summing blocks. The closed-loop model is obtained by

$$\text{CL}_1 = \mathcal{F}_l\left(\text{nominal\_augment\_lateral\_system}, K_{\text{plant}}\right), \tag{31}$$

where $\mathcal{F}_l(\cdot,\cdot)$ denotes the lower linear fractional transformation. This model has physical external inputs

$$[r_y,\ r_\psi,\ v_w]^T \tag{32}$$

and analysis outputs

$$[y_{\text{rig}},\ \psi,\ v_y,\ r,\ e_y,\ e_\psi]^T. \tag{33}$$

This separation is important because the tracking and disturbance weights are used for synthesis, but they are not interpreted as physical vehicle dynamics in the final analysis.

5 Weighting Design

5.1 Lateral Tracking Weight

The lateral tracking weight is

$$W_{a_y}(s) = \frac{1}{e_{y,\max}} \frac{\omega_b}{s+\omega_b}, \tag{34}$$

where

$$e_{y,\max} = 0.18 \text{ m}, \qquad \omega_b = 2\pi(0.5) \text{ rad/s}. \tag{35}$$

At low frequency,

$$|W_{a_y}(0)| = \frac{1}{0.18} = 5.56 \text{ m}^{-1}. \tag{36}$$

Thus the design asks the controller to keep low-frequency lateral error below approximately 0.18 m. Above the selected bandwidth, the weight rolls off and high-frequency tracking error is penalized less strongly.

5.2 Yaw Tracking Weight

The yaw tracking weight is defined as

$$W_{a_\psi}(s) = \frac{1}{e_{\psi,\max}} \frac{\omega_b}{s+\omega_b}, \qquad e_{\psi,\max} = 0.3 \text{ rad}. \tag{37}$$

However, this weight is not currently active in synthesis because the current generalized-plant output vector does not include $W_{a_\psi} e_\psi$. If heading tracking is a formal design requirement, the output vector should include both weighted errors.

5.3 Steering Command Weight

The steering command shaping block is

$$W_{\text{cmd}}(s) = \frac{0.5}{(0.05/\pi)s + 1}. \tag{38}$$

This maps the normalized controller command to the physical steering command. Its DC gain is 0.5 rad and its pole is approximately 62.8 rad/s. This block shapes the steering path, but it is not a hard saturation model.

5.4 Side-Wind Weight

The side-wind disturbance weight is

$$W_{\text{wind}}(s) = \frac{4.9}{s + 1}. \tag{39}$$

The low-frequency gain represents a wind disturbance on the order of several meters per second, while the pole at 1 rad/s emphasizes slow and step-like wind changes. If the expected operating scenario is a larger sudden gust, the DC gain and pole of W_{wind} should be selected to match that disturbance model.

6 Simulation and Robustness Results

Running the current MATLAB script gives the open-loop poles

$$p = [0,\ 0,\ -22.1741,\ -39.3899]\,. \tag{40}$$

The two poles at the origin come from the lateral-position and yaw-angle integrators. The latest H_∞ synthesis result is

$$\gamma = 0.4641. \tag{41}$$

Since $\gamma < 1$, the active weighted lateral-error objective is feasible for the shaped synthesis plant.

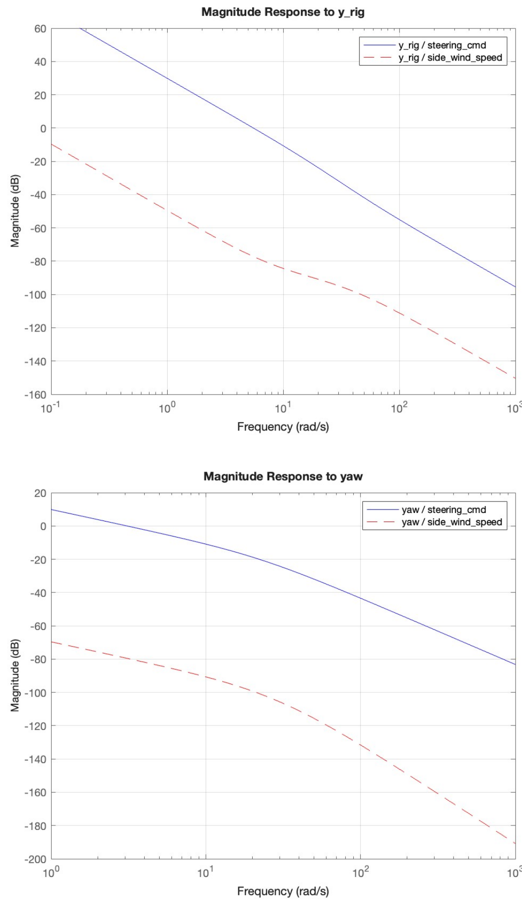

Figure 2: Open-loop plant magnitude responses from steering command and side-wind speed to lateral position and yaw. Steering has much larger authority than the side-wind channel, while the side-wind-to-yaw transfer is very small because wind force is applied at the CG.

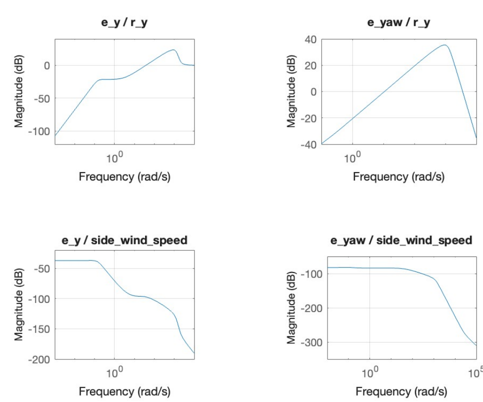

Figure 3: Nominal closed-loop tracking-error magnitude responses without and with the measurement-noise channels. The two plots are visually identical here because the current noise weights are set to zero in the MATLAB script.

Table 1: Current time-domain simulation results

Case	Reference	Wind	Peak $\lvert e_y \rvert$
No side wind	default S path, ± 2 m	0 m/s	0.264 m
Wind step	0 m	8 m/s at 5 s	0.246 m

The two cases in Table 1 should not be compared as a direct wind/no-wind performance pair because they use different references. The no-wind case evaluates nominal trajectory tracking, while the wind-step case isolates disturbance rejection with zero lateral reference. A fair comparison should use the same reference trajectory and only change the side-wind input.

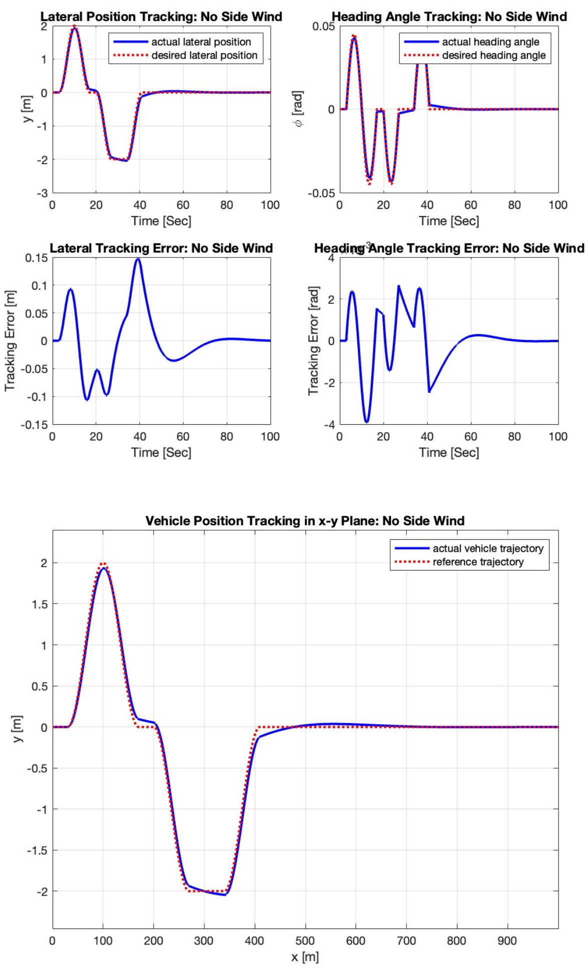

Figure 4: Nominal closed-loop tracking of the S-shaped lateral reference with no side wind. The lateral position follows the ± 2 m path with small transient offset, and the reconstructed x-y trajectory confirms that the physical vehicle path follows the desired lane-change shape.

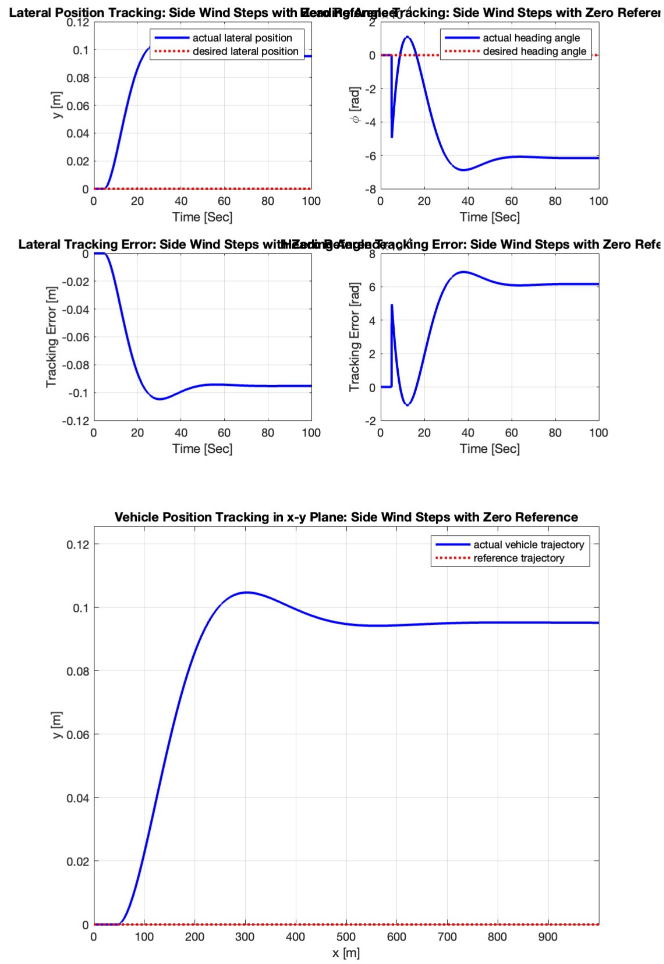

Figure 5: Nominal closed-loop response to an 8 m/s side-wind step with zero lateral reference. The controller limits the lateral displacement to roughly 0.1 m, but the yaw response is large because yaw tracking is not part of the active synthesis objective.

6.1 Parametric Uncertainty

Robustness is evaluated by replacing the nominal mass and tire cornering stiffnesses with uncertain real parameters. The uncertainty set used for the reported MATLAB run is

$$m \in [-5\%, +15\%], \qquad C_{af}, C_{ar} \in [-10\%, 0\%], \tag{42}$$

relative to their nominal values. The nominal H_∞ controller is then closed around the uncertain physical plant and tested using robust stability, robust performance, and sampled closed-loop simulations.

Table 2: Robustness and μ-synthesis command results

Quantity	Result
H_∞ synthesis γ	0.4641
Robust stability margin, nominal controller	[0.2129, 0.2133]
Robust performance margin at $\gamma = 1$	[0.1320, 0.1323]
Best μ-synthesis robust performance	1.03
Final `musyn` robust performance	1.0258
μ controller nominal closed-loop stable	true

The robust stability margin is below one, so the nominal H_∞ controller is not certified to stabilize the full uncertainty set. A margin of about 0.213 means the closed loop can tolerate only about 21% of the modeled uncertainty size before worst-case instability may occur. The robust performance margin is even smaller, which means the weighted tracking-performance target is lost before the worst-case stability boundary is reached.

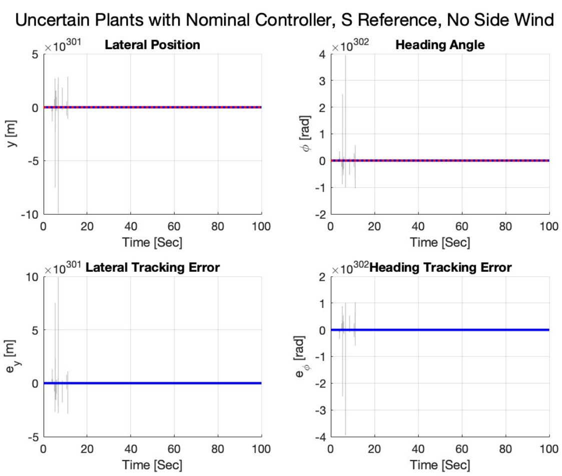

Figure 6: Sampled uncertain plants with the nominal H_∞ controller for the S-shaped reference and zero side wind. Several sampled responses diverge, which is consistent with the robust stability margin being far below one.

7 μ-Synthesis Controller Design

Because the nominal H_∞ controller is not certified for the full uncertainty set, a structured robust controller is also designed using MATLAB `musyn`. The same uncertain generalized plant, input/output ordering, single controller measurement e_y, and single controller command δ_{in} are used. The implemented physical controller is

$$K_{\mu,\text{plant}}(s) = W_{\text{cmd}}(s)K_\mu(s), \tag{43}$$

so the controller output is converted to the physical steering command before closing the loop around the vehicle plant.

`musyn` applies D-K iteration. In each iteration, the K step synthesizes a controller for the scaled problem, then the D step fits frequency-dependent scalings that upper-bound the structured singular value. The command output in Table 3 shows that the peak μ value decreases from 6.358 in the first iteration to 1.026 by the ninth iteration.

Table 3: D-K iteration summary from `musyn`

Iter.	K step	Peak μ	D fit	D order
1	2.695×10^6	6.358	6.423	46
2	1.638	1.504	1.514	20
3	1.205	1.205	1.212	26
4	1.097	1.097	1.100	28
5	1.063	1.063	1.065	30
6	1.050	1.049	1.051	32
7	1.038	1.036	1.038	32
8	1.034	1.033	1.037	32
9	1.026	1.026	1.026	32

The best achieved robust performance reported by `musyn` is about 1.03, and the final returned value is 1.0258. This is a substantial improvement over the nominal H_∞ robust-performance margin, but the value is still slightly above one. Therefore the μ-synthesis controller improves the robust design, but it still does not fully certify robust performance over the complete modeled uncertainty set.

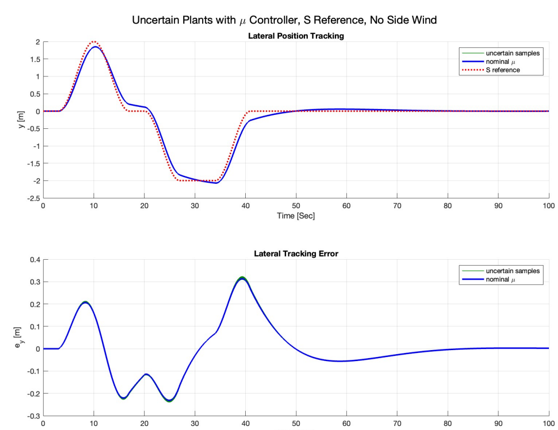

Figure 7: Sampled uncertain plants with the μ-synthesis controller for the S-shaped reference and zero side wind. The green responses show the sampled uncertain closed-loop plants, the blue response is the nominal μ controller response, and the red dotted curve is the reference trajectory.

8 Discussion

The current design achieves $\gamma < 1$ for the active nominal lateral-position tracking objective, but the robust analysis shows that nominal H_∞ performance is not enough to guarantee behavior over the modeled mass and tire uncertainty. The sampled uncertain responses in Fig. 6 show why the robust margin matters: a controller that tracks well on the nominal plant can still become unstable for admissible uncertain plants. The μ-synthesis sampled responses in Fig. 7 show that the structured robust design moves the closed-loop behavior closer to robust feasibility, but the achieved robust performance is still above one.

Several limitations remain. First, the yaw performance weight is defined but not active in the H_∞ or μ-synthesis performance output. Second, the steering command block shapes steering authority but does not enforce hard angle or rate limits. Third, the wind model is a local linearization around a selected operating wind speed, so very small or very large wind speeds may differ from the nonlinear aerodynamic force law. Finally, the weight W_{a_y} specifies an absolute error target in meters, not a percentage error target. If percentage tracking error is required, the error channel should be normalized by a chosen reference scale.

9 Conclusion

This project constructs a complete MATLAB H_∞ lateral steering design workflow for a constant-speed linear bicycle model. The final code separates the physical plant, the weighted synthesis plant, and the unweighted LFT analysis plant. This makes the design easier to interpret: the weights guide controller synthesis, while the final closed-loop simulations evaluate the controller on the original vehicle dynamics. The current controller satisfies the active nominal weighted lateral-error objective, but robust analysis shows that the full modeled uncertainty range is not certified. The μ-synthesis controller improves the robust objective but still gives a robust-performance value slightly above one. Future work should activate yaw tracking in the synthesis objective, verify steering saturation and rate limits, and either relax the tracking weight or reduce the uncertainty set to obtain a certified robust design.

References

[1] R. C. Rafaila and G. Livint, "H-infinity control of automatic vehicle steering," in *2016 International Conference and Exposition on Electrical and Power Engineering (EPE)*, Iasi, Romania, 2016, pp. 031–036, doi: 10.1109/ICEPE.2016.7781297.